

AUTO TICKET CLASSIFICATION USING FLOW DESIGNER

Project Documentation

1. Introduction

Project Title: Auto Ticket Classification using Flow Designer

Team Members:

- Jagathees P – Team Leader
- Logeshwaran S
- Nagarajan S J
- Dineshkumar Narayanan
- Sathishkumar Harikrishnan

2. Project Overview

Purpose

The purpose of this project is to automate the classification of service tickets using ServiceNow Flow Designer. The system reduces manual classification effort by automatically identifying the appropriate category and subcategory for a ticket and supporting automated routing and processing.

Features

- Automatic ticket classification.
- Category and subcategory selection.
- Automated ticket routing and assignment.
- Flow Designer based workflow automation.
- Custom ServiceNow ticket table.
- Automatic flow execution when a ticket is created.
- REST API support for interacting with ticket records.
- Validation of category and subcategory relationships.

3. Architecture

Backend

The backend automation is implemented using ServiceNow Flow Designer. When a new ticket record is created, the configured flow is triggered, reads the ticket information, determines the appropriate classification, updates the record, and can route or notify the relevant support group.

Flow sequence: Ticket Creation → Flow Trigger → Read Ticket Details → Identify Category → Identify Subcategory → Update Ticket → Route/Notify Support Team

Database

Ticket records are primarily stored in the ServiceNow platform database. The project architecture can also interact with MySQL or MongoDB where an external application requires additional storage such as user information, ticket data, classification history, logs, or application-specific records.

Field	Description
Ticket ID	Unique ticket number
Short Description	Brief description of the issue
Description	Complete issue details
Category	Main ticket category
Subcategory	Specific issue type
Priority	Ticket priority
Status	Current ticket status
Assigned Group	Responsible support team
Created Date	Ticket creation date

4. Setup Instructions

Prerequisites

- Active ServiceNow Personal Developer Instance (PDI).
- System Administrator privileges.
- admin role to preview and commit Update Sets.
- Internet connection and a supported web browser.
- Node.js and npm if the separate frontend/backend application is used.

Installation

1. Log in to the ServiceNow Personal Developer Instance with administrator privileges.
2. Navigate to System Update Sets → Retrieved Update Sets.
3. Import the provided Update Set XML file into the target instance.
4. Open the imported Update Set and select Preview Update Set.
5. Review and resolve any conflicts or validation errors.
6. After successful validation, select Commit Update Set.
7. Verify the custom table, fields, category/subcategory configuration, and Flow Designer flow.
8. Create a test ticket and confirm that the automation executes correctly.

5. Folder Structure

Business Use Case

Organizations receive many support tickets and manual classification can be time-consuming and inconsistent. Automating classification improves processing speed and reduces repetitive work.

System Architecture

The solution contains a custom ticket table, category and subcategory fields, category/subcategory dependencies, Flow Designer triggers, actions, and ticket assignment or routing logic.

Suggested Project Structure

Auto-Ticket-Classification/

- client/
- ■■■■ src/
- ■■■■ public/
- ■■■■ package.json
- server/
- ■■■■ routes/
- ■■■■ controllers/
- ■■■■ models/
- ■■■■ package.json
- servicenow/
- ■■■■ update-set.xml
- ■■■■ flows/
- ■■■■ tables/
- screenshots/
- README.md

Test Scenarios

Testing covers new ticket creation, classification, category/subcategory validation, multiple ticket processing, API responses, and unauthorized access handling.

6. Running the Application

Frontend

```
cd client
npm install
npm start
```

Backend

```
cd server
npm install
npm start
```

The actual ports depend on the project's configuration.

7. API Documentation

Overview

ServiceNow provides REST APIs that can be used to interact with custom tables and ticket records. These APIs can create, retrieve, and update records and can work with the configured automation when records are created or updated.

Method	Purpose
GET	Retrieve ticket records
POST	Create a new ticket
PUT/PATCH	Update an existing ticket
DELETE	Remove a record when permitted

8. Authentication

Authentication and authorization are handled through ServiceNow security mechanisms. Depending on the configured integration, REST API access may use Basic Authentication, OAuth 2.0, or session-based authentication. ServiceNow roles and Access Control Lists (ACLs) determine which users or integrations can read or modify records. Credentials, tokens, and secrets should not be hard-coded in application source code.

9. User Interface

- Login or dashboard screen.
- Ticket creation screen.
- Ticket details and classification screen.
- Automatically assigned category and subcategory.
- Flow Designer configuration screen.
- Category/subcategory configuration screen.
- API response or integration screen.

Screenshots or GIFs of these screens should be inserted in the final report to demonstrate the working application.

10. Testing

The testing strategy combines functional, integration, API, Flow Designer, database, user-interface, and error-handling tests.

Test Case	Input/Action	Expected Result
TC01	Create a new ticket	Flow is triggered
TC02	Enter valid ticket description	Correct category is assigned
TC03	Select/derive category	Appropriate subcategory is available
TC04	Submit invalid/incomplete data	Validation or error is handled
TC05	Create multiple tickets	Tickets are processed independently
TC06	Send REST API request	Correct API response is returned
TC07	Unauthorized API request	Access is denied

11. Screenshots or Demo

The final project demonstration should include screenshots of the ServiceNow dashboard, custom ticket table, ticket creation, Flow Designer workflow, automatic classification, and final ticket result. A hosted demo link can also be added if available.

- Screenshot 1 – ServiceNow Dashboard
- Screenshot 2 – Custom Ticket Table
- Screenshot 3 – Ticket Creation
- Screenshot 4 – Flow Designer Workflow
- Screenshot 5 – Automatic Category Assignment
- Screenshot 6 – Automatic Subcategory Assignment
- Screenshot 7 – Final Classified Ticket
- Screenshot 8 – REST API Response

12. Known Issues

- Classification may not work when required ticket fields are missing.
- Incorrect category/subcategory configuration can affect classification.
- Flow execution may fail because of configuration errors.
- API authentication problems can prevent external applications from accessing ServiceNow.
- Network connectivity can affect API communication.
- Large ticket volumes may require performance optimization.
- Changes to ServiceNow tables or fields may require corresponding Flow updates.

13. Future Enhancements

- AI/ML-based ticket classification.
- Natural Language Processing for ticket descriptions.
- Automatic priority prediction.
- Automatic assignment to support teams.
- Email and notification integration.
- Real-time ticket monitoring dashboard.
- Ticket classification analytics and reporting.
- Automatic duplicate-ticket detection.
- Multilingual ticket classification.
- Integration with external databases and applications.
- Automatic SLA monitoring.
- Chatbot-based ticket creation.

- Use of historical ticket data to improve classification accuracy.

Conclusion

Auto Ticket Classification using Flow Designer provides an automated approach for managing and classifying service tickets in ServiceNow. By combining Flow Designer, custom tables, automated triggers, and REST APIs, the solution reduces manual effort and improves consistency in ticket processing. Future integration of AI and Machine Learning can further improve classification based on ticket descriptions and historical data.